Diploma in Engineering and Technology

# Applied Physics Lab Handbook

Complete professional laboratory textbook for Course Code 2006. This lesson is built from the supplied official syllabus and converts the experiment list into a practical learning, revision, viva and exam-preparation handbook.

Save as PDF

COURSE CODE
**2006**

TITLE
**Applied Physics Lab**

SEMESTER
**1 & 2**

CREDITS
**1**

CATEGORY
**Basic Science**

PERIODS
**2/week**

SEMESTER PERIODS
**30 + 30**

SCHEME
**Revision 2021**

Engineering lab flow

## Measure → Correct → Calculate → Verify → Report

Every practical answer must show aim, apparatus, principle, formula, diagram/circuit, procedure, observation table, calculation, result, precautions and common errors.

**20** experiments | **4** CO modules | **12 minimum** experiments to perform

__SVG_MEASURE__

# Course Overview

Applied Physics Lab develops measurement discipline, apparatus handling confidence and engineering interpretation of physics experiments.

### Objectives from supplied syllabus

- Supplement factual physics knowledge by first-hand manipulation of apparatus.
- Develop scientific temper for applying physics principles to engineering and technology problems.
- Build confidence in equipment handling and measurement skills.

### Prerequisite

Basic knowledge in Physics from Secondary School.

> □ lab subject-□ theory □□□□□□ □□□□□□□□□□. Instrument □□□□□□□□□, least count □□□□□□□□□□, graph □□□□□□□□□, result unit □□□□□ □□□□□□ practice □□□□.

### Table of Contents

### Official source information

| Item | Status | Value from supplied syllabus/source |
|---|---|---|
| Course code/title | Specified | 2006 - Applied Physics Lab |
| Programme | Specified | Diploma in Engineering and Technology |
| Semester | Specified | 1 & 2 |
| Credits/category | Specified | 1 credit, Basic Science |
| Periods | Specified | 2 per week; 30 + 30 per semester |
| CO-PO mapping | Specified | CO1, CO2, CO3 and CO4 mapped to PO1 at level 3. Other PO values are blank/not specified. |
| Exam blueprint | Limited | Lab Exam I and II are mentioned; detailed written question-paper pattern is not specified in supplied source. Model paper here is a fresh practice paper. |

## Redesigned Syllabus Roadmap

Not a copied syllabus table. This roadmap tells what to draw, calculate, explain and avoid in each module.

**Module 1 • 12 hours • CO1**

### Measurement Tools and Accuracy

**Outcome:** Select appropriate measuring tools and make measurements with accuracy and precision.

**Core topics:** Vernier calipers, Screw gauge, Spherometer, Mercury thermometer, Least count, Zero error, Unit conversion

**What to draw:** Vernier jaws and scales, Screw gauge pitch/head scale, Spherometer legs and screw, Temperature scale conversion

**What to calculate:** least count, zero correction, volume, diameter/thickness, radius of curvature, temperature conversion

**Field relevance:** Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

**Module 2 • 23 hours • CO2**

### Mechanics and Properties of Matter

**Outcome:** Apply and illustrate mechanics and properties of matter through experiments.

**Core topics:** Hooke law, Stoke law, Parallelogram law, Moment bar, U-tube, Flywheel, Pendulum, Resonance column, Cantilever

**What to draw:** spring graph, terminal velocity setup, vector force diagram, moment bar, U-tube levels, flywheel, pendulum, resonance column, cantilever

**What to calculate:** spring constant, viscosity, unknown mass, moment balance, relative density, moment of inertia, g, sound velocity, time period

**Field relevance:** Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

**Module 3 • 6 hours • CO3**

### Ray Optics Experiments

**Outcome:** Experiment with lens, prism and glass slab to realize basic laws of ray optics.

**Core topics:** Convex lens u-v method, Lens power, Refraction, Snell law, Ray tracing

**What to draw:** u-v setup, convex lens ray diagram, glass slab/prism ray diagram

**What to calculate:** focal length, power, sin i/sin r, refractive index

**Field relevance:** Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

**Module 4 • 15 hours • CO4**

### Electrical and Semiconductor Characteristics

**Outcome:** Use V-I characteristics of conductors and semiconductors to determine resistance of materials.

**Core topics:** Ohm law, series/parallel resistance, diode V-I, potentiometer, meter bridge, specific resistance

**What to draw:** ammeter-voltmeter circuit, series/parallel circuits, diode V-I graph, potentiometer circuit, meter bridge

**What to calculate:** R=V/I, equivalent resistance, knee voltage, internal resistance, specific resistance

**Field relevance:** Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

Module 1

**Module 1 • CO1 • 12 hours**

## Measurement Tools and Accuracy

Select appropriate measuring tools and make measurements with accuracy and precision.

### Chapter introduction

This chapter converts the laboratory outcome into student-friendly experiments. Study principle first, then procedure, readings, formula, result and precautions.

### Module checklist

- Core topics: Vernier calipers, Screw gauge, Spherometer, Mercury thermometer, Least count, Zero error, Unit conversion
- Draw: Vernier jaws and scales, Screw gauge pitch/head scale, Spherometer legs and screw, Temperature scale conversion
- Calculate: least count, zero correction, volume, diameter/thickness, radius of curvature, temperature conversion
- Common mistake: skipping unit, zero correction, graph scale or polarity check.

Experiment 1

## Vernier Calipers

**Aim:** Determine volume of spherical/cylindrical body and test tube using Vernier calipers.

**Apparatus:** Vernier calipers, sphere/cylinder, test tube.

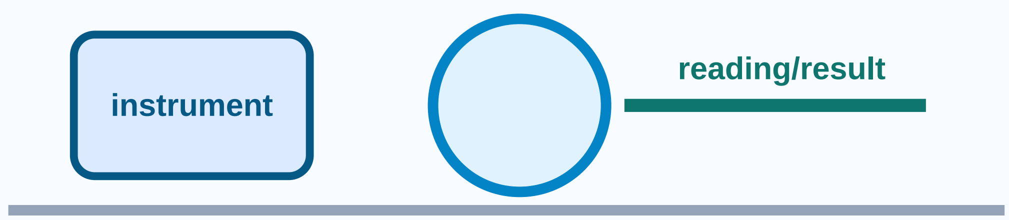
**Principle:** Vernier scale improves length measurement; least count is the smallest measurable length.

**Formula/rule:** LC = 1 MSD - 1 VSD; sphere V = 4/3 pi r^3; cylinder V = pi r^2 h

### Procedure

1. Check zero error and zero correction.
2. Hold object gently between jaws.
3. Read main scale just before Vernier zero.
4. Find coinciding Vernier division.
5. Calculate corrected dimension and volume.

### Vernier Calipers experiment diagram



**Observation table columns:** Object | MSR | Vernier coincidence | LC | Corrected dimension | Volume

**Precautions:** Avoid parallax.; Do not overtighten jaws.; Use consistent units.

**Common fault:** Ignoring zero correction gives wrong result.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 2

## Screw Gauge

**Aim:** Determine diameter of wire/ball and thickness of glass plate.

**Apparatus:** Screw gauge, wire, ball, glass plate.

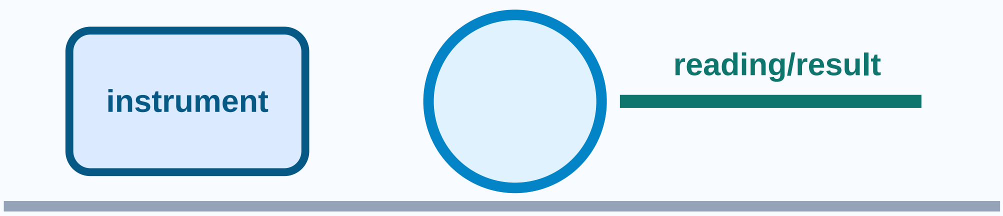
**Principle:** A screw converts rotation into small linear movement for fine measurement.

**Formula/rule:** LC = pitch / circular divisions; reading = PSR + CSR x LC +/- zero correction

### Procedure

1. Find pitch and least count.
2. Check zero error.
3. Place object between anvil and spindle.
4. Use ratchet gently.
5. Take readings at different positions.

### Screw Gauge experiment diagram



**Observation table columns:** Trial | PSR | CSR | LC | Zero correction | Corrected reading

**Precautions:** Use ratchet only.; Keep wire perpendicular.; Avoid compression.

**Common fault:** Overtightening makes reading smaller.

**Malayalam support:** Experiment result □□□□□□□□ □□□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□□ □□□□□□□□□□□□□□. Reading □□□□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□□ mistakes □□□□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 3

## Spherometer

**Aim:** Determine radius of curvature of spherical surface.

**Apparatus:** Spherometer, spherical surface, plane glass, scale.

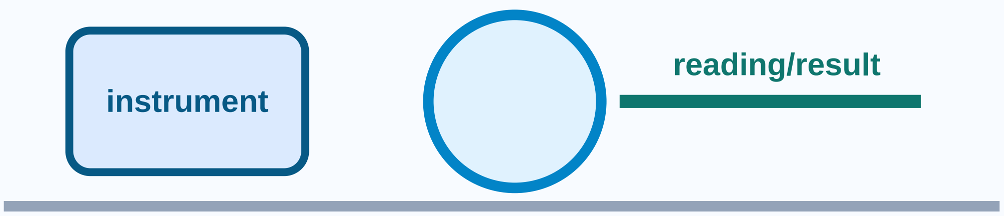
**Principle:** Sagitta h is measured with central screw and radius is calculated from leg spacing.

**Formula/rule:** R = l^2/(6h) + h/2

### Procedure

1. Find least count.
2. Take plane glass reading.
3. Take spherical surface reading.
4. Measure distances between legs.
5. Calculate h and R.

**Spherometer experiment diagram**



**Observation table columns:** l1 | l2 | l3 | mean l | h | R

**Precautions:** All legs must touch surface.; Do not drag on optical surface.; Use correct sign.

**Common fault:** Wrong sign of h is common.

**Malayalam support:** Experiment result ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯ least count, zero error, unit conversion, observation neatness ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯. Reading ▯▯▯▯▯▯▯▯▯▯▯ parallax, loose contact, overload ▯▯▯▯▯▯▯ mistakes ▯▯▯▯▯▯▯▯▯▯.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 4

## Mercury Thermometer

**Aim:** Measure room and hot bath temperature and convert scales.

**Apparatus:** Mercury thermometer, hot bath, beaker, stirrer.

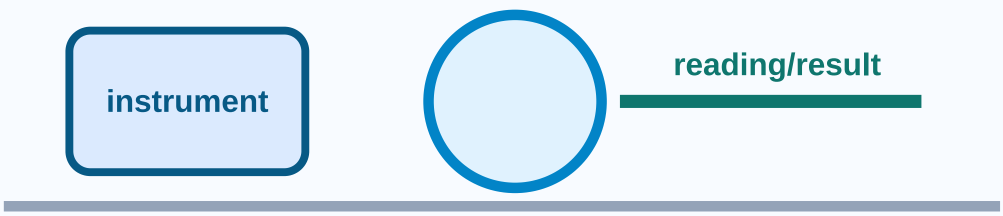
**Principle:** Mercury expansion indicates temperature.

**Formula/rule:** $K = C + 273$; $F = 9C/5 + 32$

### Procedure

1. Check range and least count.
2. Wait for stable reading.
3. Immerse bulb without touching bottom.
4. Read at eye level.
5. Convert scales.

### Mercury Thermometer experiment diagram



**Observation table columns:** Condition | Celsius | Kelvin | Fahrenheit | Remarks

**Precautions:** Do not hold bulb.; Do not hit glass.; Read at eye level.

**Common fault:** Bulb touching beaker wall changes reading.

**Malayalam support:** Experiment result □□□□□□□□ □□□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□□ mistakes □□□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

## Module 1 Expected Questions

**Vernier Calipers:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Screw Gauge:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Spherometer:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Mercury Thermometer:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

Module 2

Module 2 • CO2 • 23 hours

# Mechanics and Properties of Matter

Apply and illustrate mechanics and properties of matter through experiments.

### Chapter introduction

This chapter converts the laboratory outcome into student-friendly experiments. Study principle first, then procedure, readings, formula, result and precautions.

### Module checklist

- Core topics: Hooke law, Stoke law, Parallelogram law, Moment bar, U-tube, Flywheel, Pendulum, Resonance column, Cantilever
- Draw: spring graph, terminal velocity setup, vector force diagram, moment bar, U-tube levels, flywheel, pendulum, resonance column, cantilever
- Calculate: spring constant, viscosity, unknown mass, moment balance, relative density, moment of inertia, g, sound velocity, time period
- Common mistake: skipping unit, zero correction, graph scale or polarity check.

**Experiment 5**

## Hooke Law

**Aim:** Verify Hooke law and find force constant of spring.

**Apparatus:** Spring, weights, pointer, scale.

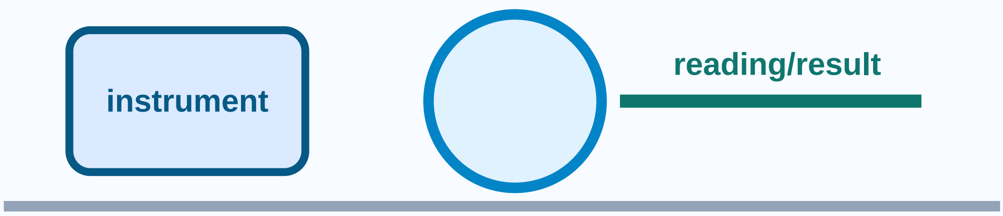
**Principle:** Within elastic limit extension is proportional to load.

**Formula/rule:** $F = kx; k = F/x$

### Procedure

1. Note initial reading.
2. Add weights gradually.
3. Record extension.
4. Plot load-extension graph.
5. Find slope k.

### Hooke Law experiment diagram



**Observation table columns:** Load | Force | Reading | Extension | k

**Precautions:** Do not cross elastic limit.; Wait until oscillation stops.; Load gently.

**Common fault:** Nonlinear graph means elastic limit crossed.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 6

## Stoke Law

**Aim:** Find viscosity using terminal velocity of a falling sphere.

**Apparatus:** Tall jar, liquid, sphere, stopwatch, scale.

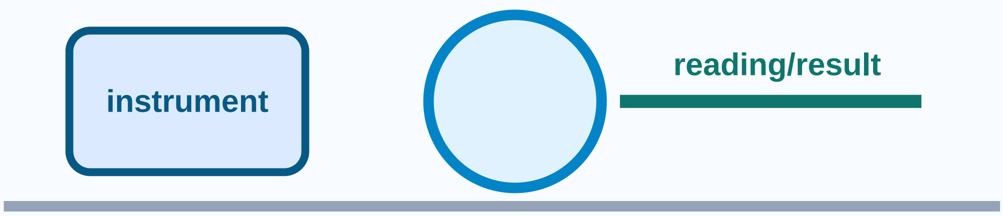
**Principle:** At terminal velocity, weight is balanced by buoyancy and viscous force.

**Formula/rule:** eta = 2r^2(rhos-rhol)g/(9v)

### Procedure

1. Measure sphere radius.
2. Mark terminal region.
3. Drop sphere centrally.
4. Time known distance.
5. Calculate velocity and viscosity.

### Stoke Law experiment diagram



**Observation table columns:** radius | distance | time | velocity | density | viscosity

**Precautions:** Avoid bubbles.; Sphere must fall centrally.; Use terminal region.

**Common fault:** Timing immediately after release is wrong.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 7

## Parallelogram Law

**Aim:** Verify force law and find unknown mass.

**Apparatus:** Force board/table, pulleys, weights.

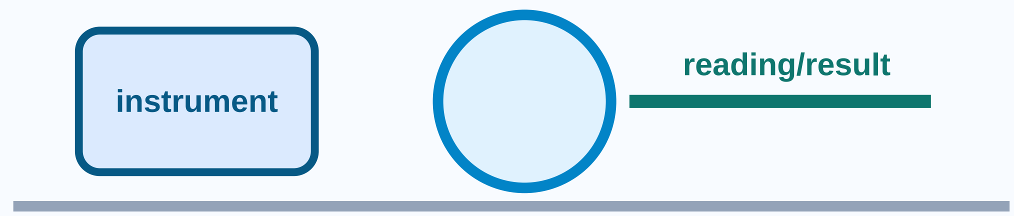
**Principle:** Two forces can be represented by adjacent sides of a parallelogram; diagonal gives resultant.

**Formula/rule:** R = sqrt(P^2+Q^2+2PQ cos theta)

### Procedure

1. Set two known weights.
2. Adjust third for equilibrium.
3. Mark string directions.
4. Draw parallelogram to scale.
5. Compare resultant with balancing weight.

**Parallelogram Law experiment diagram**



**Observation table columns:** P | Q | theta | R | balancing mass

**Precautions:** Pulleys friction-free.; Strings must meet at one point.; Use correct scale.

**Common fault:** Bad scale drawing gives wrong mass.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 8

## Moment Bar

**Aim:** Find mass using principle of moments.

**Apparatus:** Moment bar, knife edge, weights.

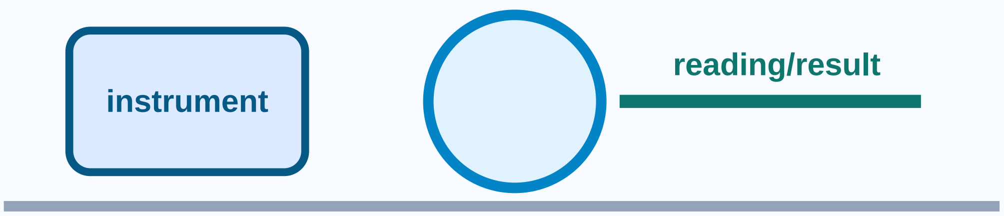
**Principle:** At equilibrium clockwise moment equals anticlockwise moment.

**Formula/rule:** W1 d1 = W2 d2

### Procedure

1. Balance empty bar.
2. Place known weight.
3. Place unknown body.
4. Adjust distances for balance.
5. Calculate mass.

### Moment Bar experiment diagram



**Observation table columns:** known mass | distance | unknown distance | unknown mass

**Precautions:** Measure from pivot.; Keep knife edge sharp.; Avoid oscillation.

**Common fault:** Measuring from wrong zero causes error.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 9

## U-tube Apparatus

**Aim:** Find relative density of liquid.

**Apparatus:** U-tube, liquids, scale.

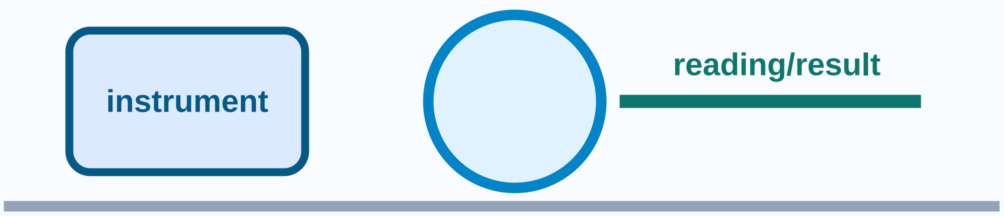
**Principle:** Pressure at same horizontal level in connected liquid columns is equal.

**Formula/rule:** rho1 h1 = rho2 h2

### Procedure

1. Pour reference liquid.
2. Add test liquid.
3. Allow levels to settle.
4. Measure heights.
5. Calculate relative density.

### U-tube Apparatus experiment diagram



**Observation table columns:** liquid | h1 | h2 | relative density

**Precautions:** Avoid bubbles.; Read meniscus correctly.; Tube vertical.

**Common fault:** Wrong meniscus level gives error.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 10

## Flywheel

**Aim:** Find moment of inertia of a flywheel.

**Apparatus:** Flywheel, axle, thread, mass, stopwatch.

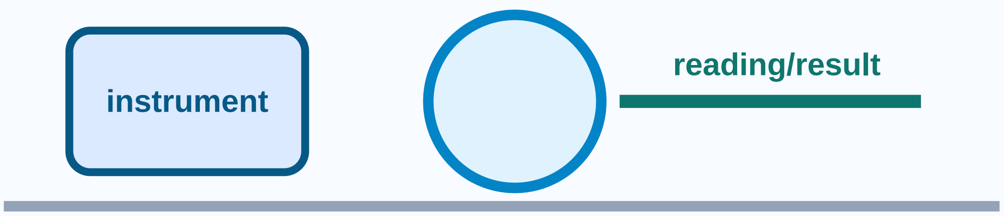
**Principle:** Falling mass energy changes into rotational kinetic energy and friction losses.

**Formula/rule:** I from energy balance using mass, height, rotations and omega

### Procedure

1. Wind thread uniformly.
2. Attach mass and release.
3. Count rotations.
4. Measure fall time and height.
5. Calculate I.

### Flywheel experiment diagram



**Observation table columns:** mass | height | time | rotations | I

**Precautions:** No thread overlap.; Count rotations carefully.; Release without push.

**Common fault:** Missing free rotations causes wrong friction correction.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 11

## Simple Pendulum

**Aim:** Find acceleration due to gravity.

**Apparatus:** Pendulum bob, thread, stand, stopwatch, scale.

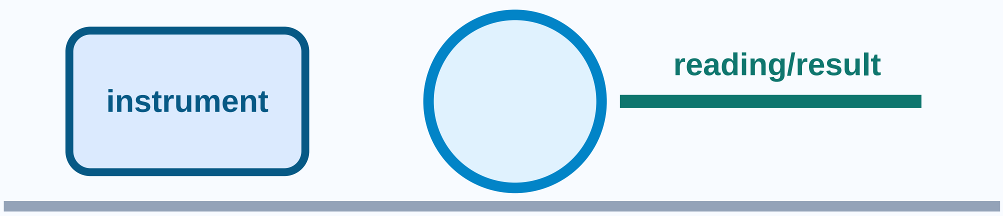
**Principle:** For small oscillations time period depends on length and g.

**Formula/rule:** T = 2 pi sqrt(l/g); g = 4 pi^2 l / T^2

### Procedure

1. Measure effective length.
2. Give small displacement.
3. Record time for 20 oscillations.
4. Repeat for different lengths.
5. Calculate T and g.

**Simple Pendulum experiment diagram**



**Observation table columns:** length | time for 20 | T | g

**Precautions:** Small amplitude.; Avoid elliptical motion.; Same reference point.

**Common fault:** Using string length instead of effective length.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 12

## Resonance Column

**Aim:** Find velocity of sound in air.

**Apparatus:** Resonance column, tuning fork, thermometer.

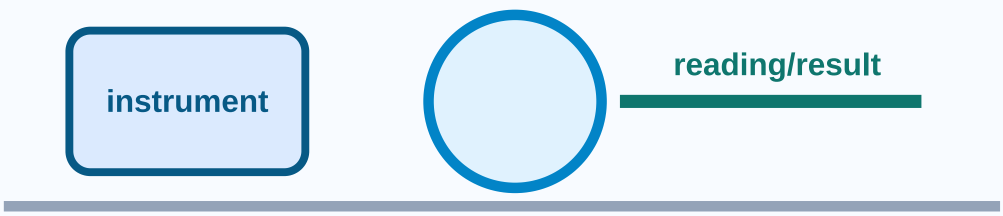
**Principle:** Air column resonates at odd-quarter wavelength lengths.

**Formula/rule:** $v = 2f(l2-l1)$

### Procedure

1. Strike fork gently.
2. Hold near tube.
3. Adjust water level.
4. Find first and second resonance.
5. Calculate velocity.

**Resonance Column experiment diagram**



**Observation table columns:** frequency | l1 | l2 | temperature | v

**Precautions:** Do not touch fork.; Use same fork position.; Read water level at eye level.

**Common fault:** Ignoring end correction can reduce accuracy.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 13

## Cantilever

**Aim:** Find and verify time period of oscillation.

**Apparatus:** Cantilever strip, load, stopwatch.

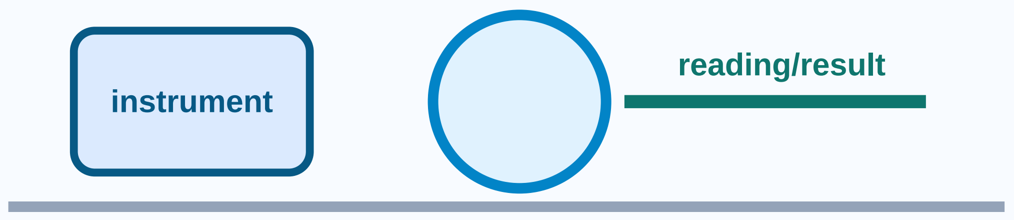
**Principle:** A loaded cantilever performs oscillation depending on load and stiffness.

**Formula/rule:** T depends on effective mass and stiffness

### Procedure

1. Clamp one end firmly.
2. Attach load at free end.
3. Displace slightly.
4. Measure time for oscillations.
5. Compare time periods.

**Cantilever experiment diagram**



**Observation table columns:** load | length | time | T

**Precautions:** Clamp rigidly.; Small oscillation only.; Avoid air disturbance.

**Common fault:** Loose clamp changes result.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

## Module 2 Expected Questions

**Hooke Law:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Stoke Law:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Parallelogram Law:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Moment Bar:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**U-tube Apparatus:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Flywheel:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Simple Pendulum:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Resonance Column:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Cantilever:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

Module 3

Module 3 • CO3 • 6 hours

# Ray Optics Experiments

Experiment with lens, prism and glass slab to realize basic laws of ray optics.

### Chapter introduction

This chapter converts the laboratory outcome into student-friendly experiments. Study principle first, then procedure, readings, formula, result and precautions.

### Module checklist

- Core topics: Convex lens u-v method, Lens power, Refraction, Snell law, Ray tracing
- Draw: u-v setup, convex lens ray diagram, glass slab/prism ray diagram
- Calculate: focal length, power, sin i/sin r, refractive index
- Common mistake: skipping unit, zero correction, graph scale or polarity check.

Experiment 14

## Convex Lens

**Aim:** Find focal length and power by u-v method.

**Apparatus:** Optical bench, convex lens, object, screen.

**Principle:** Lens equation relates object distance, image distance and focal length.

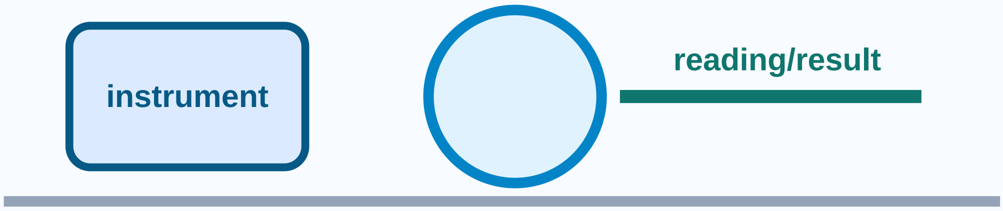
**Formula/rule:** $1/f = 1/u + 1/v$; $P = 1/f$ metre

**Procedure**

1. Align object lens and screen.
2. Find sharp image.
3. Record u and v.

4 Repeat.

5 Calculate f and power.




Convex Lens experiment diagram

**Observation table columns:** u | v | f | P

**Precautions:** Same height.; Use sharp image.; Avoid parallax.

**Common fault:** Sign convention confusion.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 15

## Snell Law

**Aim:** Verify Snell law using glass slab/prism.

**Apparatus:** Glass slab/prism, pins, protractor, scale.

**Principle:** For two media, sin i / sin r is constant.

**Formula/rule:** mu = sin i / sin r

### Procedure

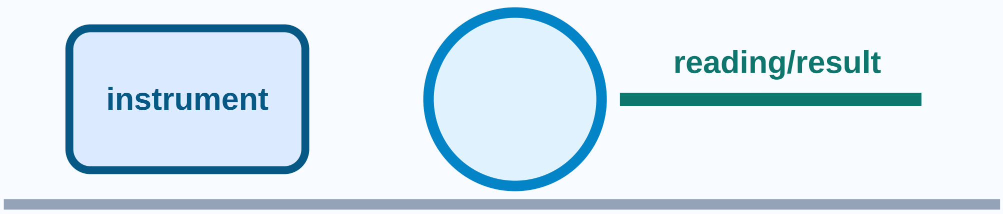
1 Trace glass outline.

2 Draw normal.

3 Fix incident ray pins.

4. Place emergent ray pins.
5. Measure i and r.

## Snell Law experiment diagram




**Observation table columns:** i | r | sin i | sin r | mu

**Precautions:** Pins vertical.; Wide pin separation.; Draw normal accurately.

**Common fault:** Wrong normal spoils ray diagram.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

## Module 3 Expected Questions

**Convex Lens:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Snell Law:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

Module 4

Module 4 • CO4 • 15 hours

# Electrical and Semiconductor Characteristics

Use V-I characteristics of conductors and semiconductors to determine resistance of materials.

## Chapter introduction

This chapter converts the laboratory outcome into student-friendly experiments. Study principle first, then procedure, readings, formula, result and precautions.

## Module checklist

- Core topics: Ohm law, series/parallel resistance, diode V-I, potentiometer, meter bridge, specific resistance
- Draw: ammeter-voltmeter circuit, series/parallel circuits, diode V-I graph, potentiometer circuit, meter bridge
- Calculate: R=V/I, equivalent resistance, knee voltage, internal resistance, specific resistance
- Common mistake: skipping unit, zero correction, graph scale or polarity check.

Experiment 16

## Ohm Law

**Aim:** Verify Ohm law by V-I graph.

**Apparatus:** DC supply, resistor, ammeter, voltmeter, rheostat.

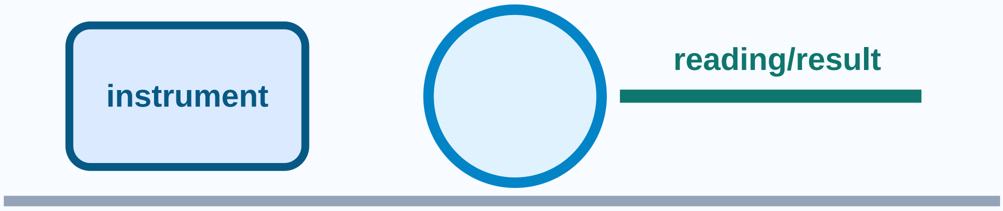
**Principle:** At constant temperature current is proportional to voltage.

**Formula/rule:** $V = IR$; $R = V/I$

### Procedure

1. Connect ammeter in series.
2. Connect voltmeter across resistor.
3. Vary rheostat.
4. Record V and I.
5. Plot graph and find slope.

## Ohm Law experiment diagram




**Observation table columns:** V | I | R | slope

**Precautions:** Check polarity.; Avoid heating.; Use proper range.

**Common fault:** Loose contacts fluctuate current.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 17

### Laws of Resistances

**Aim:** Verify series and parallel laws.

**Apparatus:** Resistors, supply, meters, wires.

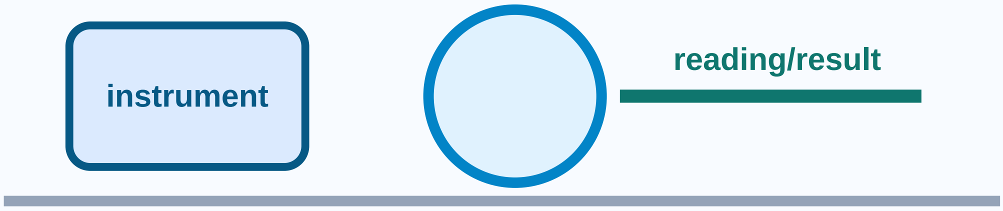
**Principle:** Series resistances add; in parallel reciprocals add.

**Formula/rule:** $R_s = R_1+R_2$; $1/R_p = 1/R_1+1/R_2$

#### Procedure

1. Measure individual resistances.
2. Connect series and observe equivalent.
3. Connect parallel and observe equivalent.
4. Compare with theory.
5. Find error.

## Laws of Resistances experiment diagram




**Observation table columns:** R1 | R2 | theory | observed | error

**Precautions:** Correct wiring.; No short circuit.; Proper meter range.

**Common fault:** Parallel branch often gets shorted.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 18

### Diode V-I Characteristics

**Aim:** Draw V-I curve and find knee voltage.

**Apparatus:** Diode, supply, resistor, milliammeter, voltmeter.

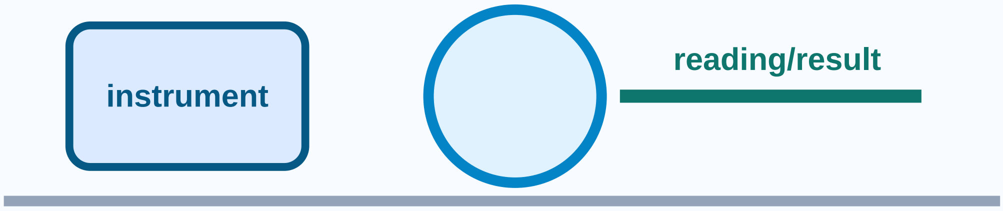
**Principle:** Diode conducts in forward bias after knee voltage and blocks in reverse bias.

**Formula/rule:** Ge knee about 0.3 V; Si knee about 0.7 V

#### Procedure

1. Connect forward bias with resistor.
2. Increase voltage slowly.
3. Record current.
4. Plot curve.
5. Identify knee voltage.

## Diode V-I Characteristics experiment diagram




**Observation table columns:** Vf | If | Vr | Ir | knee voltage

**Precautions:** Use current limiting resistor.; Do not exceed rating.; Check polarity.

**Common fault:** Direct connection can destroy diode.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 19

### Potentiometer

**Aim:** Find internal resistance/comparison of emfs.

**Apparatus:** Potentiometer, cells, galvanometer, jockey.

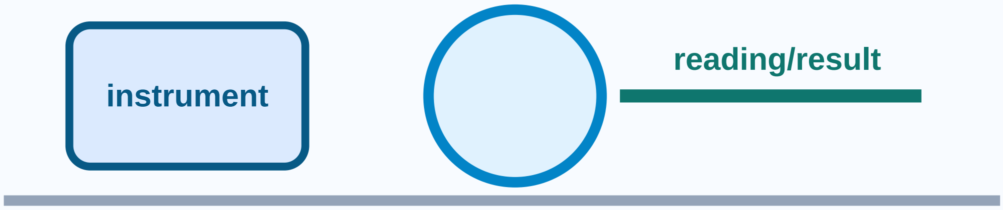
**Principle:** Potential drop along uniform wire is proportional to length at balance.

**Formula/rule:** $E_1/E_2 = l_1/l_2$; $r = R(l_1-l_2)/l_2$

#### Procedure

1. Set primary circuit.
2. Find balance length open cell.
3. Connect external R and find new length.
4. Calculate r.
5. Compare emfs using lengths.

### Potentiometer experiment diagram




**Observation table columns:** cell | l1 | l2 | R | r/emf ratio

**Precautions:** Do not drag jockey.; Driver emf must be higher.; Check polarity.

**Common fault:** Wrong polarity gives no balance.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

Experiment 20

## Meter Bridge

**Aim:** Find specific resistance of wire.

**Apparatus:** Meter bridge, resistance box, galvanometer, wire.

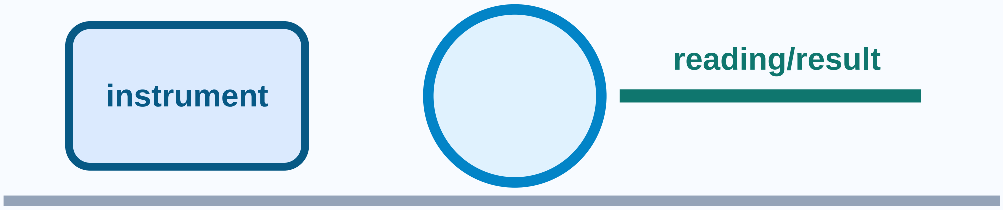
**Principle:** At Wheatstone bridge balance, ratio of resistances equals ratio of wire lengths.

**Formula/rule:** $X/R = l/(100-l)$; $\rho = RA/L$

### Procedure

1. Connect unknown and resistance box.
2. Find null point.
3. Interchange gaps.
4. Measure wire length and diameter.
5. Calculate rho.

## Meter Bridge experiment diagram




**Observation table columns:** R | I | X | L | d | rho

**Precautions:** Do not press jockey hard.; Balance near middle.; Remove key when idle.

**Common fault:** Oxidised contact causes wrong balance.

**Malayalam support:** Experiment result □□□□□□□ □□□□□□□ least count, zero error, unit conversion, observation neatness □□□□□□ □□□□□□□□□□□□. Reading □□□□□□□□□□□□□ parallax, loose contact, overload □□□□□□□ mistakes □□□□□□□□□□□.

**Exam answer format:** Aim → apparatus → labelled diagram/circuit → principle → formula → procedure → observation table → calculation → result with unit → precautions.

## Module 4 Expected Questions

**Ohm Law:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Laws of Resistances:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Diode V-I Characteristics:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Potentiometer:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

**Meter Bridge:** Write aim, draw labelled setup/circuit, explain principle, derive/apply formula, prepare observation table, list precautions and mention one troubleshooting fault.

## Formula / Rule Bank

Formula, variable meaning, unit and use. For lab exams, write formula before substitution and final result with unit.

---

### Vernier LC

$$LC = 1\ MSD - 1\ VSD$$

**Unit:** mm/cm

**Use:** smallest Vernier reading

**Example:** Substitute readings with units and write final result correctly.

### Screw gauge LC

$$LC = \text{pitch} / \text{circular divisions}$$

**Unit:** mm

**Use:** fine thickness/diameter

**Example:** Substitute readings with units and write final result correctly.

### Sphere volume

$$V = 4/3\ \pi\ r^3$$

**Unit:** $cm^3$

**Use:** sphere volume

**Example:** Substitute readings with units and write final result correctly.

### Cylinder volume

$$V = \pi\ r^2\ h$$

**Unit:** $cm^3$

**Use:** cylinder/test tube

**Example:** Substitute readings with units and write final result correctly.

### Spherometer

$$R = l^2/(6h)+h/2$$

**Unit:** cm

**Use:** radius of curvature

**Example:** Substitute readings with units and write final result correctly.

### Temperature

```
K=C+273; F=9C/5+32
```

**Unit:** K/F

**Use:** scale conversion

**Example:** Substitute readings with units and write final result correctly.

### Hooke law

```
F=kx
```

**Unit:** N/m

**Use:** spring constant

**Example:** Substitute readings with units and write final result correctly.

### Stoke law

```
eta=2r^2(rhos-rhol)g/(9v)
```

**Unit:** Pa s

**Use:** viscosity

**Example:** Substitute readings with units and write final result correctly.

### Moment

```
W1d1=W2d2
```

**Unit:** N m

**Use:** moment bar

**Example:** Substitute readings with units and write final result correctly.

### Pendulum

```
g=4pi^2l/T^2
```

**Unit:** m/s^2

**Use:** gravity

**Example:** Substitute readings with units and write final result correctly.

### Sound

$$v=2f(l2-l1)$$

**Unit:** m/s

**Use:** resonance column

**Example:** Substitute readings with units and write final result correctly.

### Lens

$$1/f=1/u+1/v$$

**Unit:** m

**Use:** focal length

**Example:** Substitute readings with units and write final result correctly.

### Lens power

$$P=1/f$$

**Unit:** D

**Use:** lens power

**Example:** Substitute readings with units and write final result correctly.

### Snell

$$mu=\sin i/\sin r$$

**Unit:** none

**Use:** refractive index

**Example:** Substitute readings with units and write final result correctly.

### Ohm

$$V=IR$$

**Unit:** ohm

**Use:** resistance

**Example:** Substitute readings with units and write final result correctly.

### Series R

$$Rs=R1+R2+...$$

**Unit:** ohm

**Use:** series circuit

**Example:** Substitute readings with units and write final result correctly.

### Parallel R

$$1/Rp=1/R1+1/R2$$

**Unit:** ohm

**Use:** parallel circuit

**Example:** Substitute readings with units and write final result correctly.

### Diode

$$Rf=DeltaV/DeltaI$$

**Unit:** ohm

**Use:** V-I curve

**Example:** Substitute readings with units and write final result correctly.

### Meter bridge

$$rho=RA/L$$

**Unit:** ohm m

**Use:** specific resistance

**Example:** Substitute readings with units and write final result correctly.

## Practice, Expected Questions and Quick Revision

Module-wise viva, short-answer, numerical, diagram and practical-style questions.

---

## Module 1: Measurement Tools and Accuracy

**Short/viva:** What is the principle of Vernier Calipers?
**Diagram:** Draw the setup/circuit.

**Numerical:** Use LC = 1 MSD - 1 VSD; sphere V = 4/3 pi r^3; cylinder V = pi r^2 h.
**Practical:** Mention precautions: Avoid parallax.; Do not overtighten jaws.; Use consistent units..

**Short/viva:** What is the principle of Screw Gauge?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use LC = pitch / circular divisions; reading = PSR + CSR x LC +/- zero correction.
**Practical:** Mention precautions: Use ratchet only.; Keep wire perpendicular.; Avoid compression..

**Short/viva:** What is the principle of Spherometer?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use R = l^2/(6h) + h/2.
**Practical:** Mention precautions: All legs must touch surface.; Do not drag on optical surface.; Use correct sign..

**Short/viva:** What is the principle of Mercury Thermometer?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use K = C + 273; F = 9C/5 + 32.
**Practical:** Mention precautions: Do not hold bulb.; Do not hit glass.; Read at eye level..

## Module 2: Mechanics and Properties of Matter

**Short/viva:** What is the principle of Hooke Law?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use F = kx; k = F/x.
**Practical:** Mention precautions: Do not cross elastic limit.; Wait until oscillation stops.; Load gently..

**Short/viva:** What is the principle of Stoke Law?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use eta = 2r^2(rhos-rhol)g/(9v).
**Practical:** Mention precautions: Avoid bubbles.; Sphere must fall centrally.; Use terminal region..

**Short/viva:** What is the principle of Parallelogram Law?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use R = sqrt(P^2+Q^2+2PQ cos theta).
**Practical:** Mention precautions: Pulleys friction-free.; Strings must meet at one point.; Use correct scale..

**Short/viva:** What is the principle of Moment Bar?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use W1 d1 = W2 d2.
**Practical:** Mention precautions: Measure from pivot.; Keep knife edge sharp.; Avoid oscillation..

**Short/viva:** What is the principle of U-tube Apparatus?
**Diagram:** Draw the setup/circuit.

**Numerical:** Use rho1 h1 = rho2 h2.
**Practical:** Mention precautions: Avoid bubbles.; Read meniscus correctly.; Tube vertical..

**Short/viva:** What is the principle of Flywheel?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use I from energy balance using mass, height, rotations and omega.
**Practical:** Mention precautions: No thread overlap.; Count rotations carefully.; Release without push..

**Short/viva:** What is the principle of Simple Pendulum?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use T = 2 pi sqrt(l/g); g = 4 pi^2 l / T^2.
**Practical:** Mention precautions: Small amplitude.; Avoid elliptical motion.; Same reference point..

**Short/viva:** What is the principle of Resonance Column?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use v = 2f(l2-l1).
**Practical:** Mention precautions: Do not touch fork.; Use same fork position.; Read water level at eye level..

**Short/viva:** What is the principle of Cantilever?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use T depends on effective mass and stiffness.
**Practical:** Mention precautions: Clamp rigidly.; Small oscillation only.; Avoid air disturbance..

## Module 3: Ray Optics Experiments

**Short/viva:** What is the principle of Convex Lens?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use 1/f = 1/u + 1/v; P = 1/f metre.
**Practical:** Mention precautions: Same height.; Use sharp image.; Avoid parallax..

**Short/viva:** What is the principle of Snell Law?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use mu = sin i / sin r.
**Practical:** Mention precautions: Pins vertical.; Wide pin separation.; Draw normal accurately..

## Module 4: Electrical and Semiconductor Characteristics

**Short/viva:** What is the principle of Ohm Law?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use V = IR; R = V/I.
**Practical:** Mention precautions: Check polarity.; Avoid heating.; Use proper range..

**Short/viva:** What is the principle of Laws of Resistances?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use Rs = R1+R2; 1/Rp = 1/R1+1/R2.
**Practical:** Mention precautions: Correct wiring.; No short circuit.; Proper meter range..

**Short/viva:** What is the principle of Diode V-I Characteristics?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use Ge knee about 0.3 V; Si knee about 0.7 V.
**Practical:** Mention precautions: Use current limiting resistor.; Do not exceed rating.; Check polarity..

**Short/viva:** What is the principle of Potentiometer?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use E1/E2 = l1/l2; r = R(l1-l2)/l2.
**Practical:** Mention precautions: Do not drag jockey.; Driver emf must be higher.; Check polarity..

**Short/viva:** What is the principle of Meter Bridge?
**Diagram:** Draw the setup/circuit.
**Numerical:** Use X/R = l/(100-l); rho = RA/L.
**Practical:** Mention precautions: Do not press jockey hard.; Balance near middle.; Remove key when idle..

## Objective Practice Answer Key

**1.** Least count means smallest measurable value. **2.** Ohm law graph slope gives resistance. **3.** Meter bridge is based on Wheatstone bridge. **4.** Snell law gives refractive index. **5.** Diode knee voltage is read from forward V-I curve.

## Fresh Model Question Paper

Prepared for practice from the supplied syllabus. It does not copy official questions.

---

**Time:** 3 Hours **Maximum marks:** 75 **Pattern:** Fresh practice paper based on lab syllabus.

### Part A - Answer all, 2 marks each

1. Define least count and zero error.

2. State Hooke law.

3. What is terminal velocity?

4. State Snell law.

5. What is diode knee voltage?

6. Write balance condition in meter bridge.

7. Mention two thermometer precautions.

8. Why is a current-limiting resistor used with diode?

## Part B - Answer any six, 5 marks each

1. Explain screw gauge diameter measurement.

2. Describe parallelogram law experiment.

3. Explain simple pendulum method to find g.

4. Describe resonance column method.

5. Explain u-v method for convex lens.

6. Verify Ohm law using V-I graph.

7. Draw diode V-I characteristics and find knee voltage.

8. Find specific resistance using meter bridge.

## Part C - Answer any four, 10 marks each

1. With diagram, explain Vernier calipers experiment for volume measurement.

2. Explain Stoke law viscosity experiment with formula and precautions.

3. Explain flywheel experiment and energy idea.

4. Verify Snell law using glass slab/prism with ray diagram.

5. Verify laws of series and parallel resistances with circuits.

6. Explain potentiometer method for emf comparison/internal resistance.

## Complete Model Answers

Answer guide with expected points, formula/diagram hints, precautions and common mistakes.

---

**General answer format**

Every model answer should contain definition/principle, labelled diagram or circuit, formula with variables, procedure, observation table, calculation, result with unit, precautions and common mistakes.

**Answer 1:** Vernier calipers: define least count, zero correction, jaws, MSR+VSRxLC, volume formula and precautions.

**Answer 2:** Screw gauge: pitch, LC, PSR+CSR x LC, ratchet use and zero correction.

**Answer 3:** Hooke law: F proportional to x, plot graph, slope gives k, stay within elastic limit.

**Answer 4:** Stoke law: terminal velocity, eta formula, central fall, avoid bubbles, use terminal region.

**Answer 5:** Optics: draw normal/rays, use lens equation or Snell ratio, avoid parallax and thick lines.

**Answer 6:** Electrical: show correct ammeter/voltmeter connection, polarity, current limiting, graph slope and loose-contact troubleshooting.

**Answer 7:** Meter bridge/potentiometer: null method, balance condition, jockey precautions and connection polarity.

## References


- Text Book of Physics for Class XI & XII (Part-I, Part-II), N.C.E.R.T., Delhi.
- Comprehensive Practical Physics, Vol. I & II, J. N. Jaiswal, Laxmi Publications (P) Ltd., New Delhi.
- Practical Physics by C. L. Arora, S. Chand & Company Ltd.


**Online resources:** Supplied source mentions e-books/e-tools/learning physics software/YouTube videos/websites etc. without specific URLs. Official references with exact URLs were not specified in supplied source; use trusted institutional resources as supplementary learning links.